

Intelligent Terms & Conditions Risk Flagging Using ML

PROJECT PROBLEM STATEMENT



SUBMITTED BY

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1 Problem Statement

Terms of Service (ToS) and Privacy Policy documents are long, dense, and written in legal language that most users do not read or understand. As a result, users routinely and unknowingly agree to clauses that waive their rights, permit extensive data sharing, allow unilateral changes to terms without notice, or limit platform liability. This gap between what users agree to and what they actually understand undermines informed consent and leaves users exposed to risks they never intended to accept.

The objective of this project is to develop and evaluate machine learning approaches for automatically classifying individual clauses from ToS and Privacy Policy documents according to the risk they pose to the user.

The project will use a clause-level dataset derived from ToSDR (Terms of Service; Didn't Read), a crowd-sourced platform that rates clauses from hundreds of real-world services. Each clause is associated with a quoted text, a topic category (e.g., data sharing, liability, termination rights), and a community-assigned rating.

The central problem addressed in this project is:

*Can machine learning models accurately classify individual ToS/Privacy Policy clauses as **risky**, **neutral**, or **safe** based on the language and structure of the clause text?*

The project will involve exploring the dataset, performing appropriate data preprocessing and exploratory data analysis, engineering and selecting relevant features, training classification models, and evaluating their performance using suitable classification metrics.

2 Goal of the Project

The primary goal of this project is to develop a reliable machine learning-based system for classifying ToS/Privacy Policy clauses as **risky**, **neutral**, or **safe**, helping users quickly identify the parts of a document that need their attention.

The specific objectives are:

1. To understand and explore the characteristics of the ToSDR clause-level dataset.
2. To perform exploratory data analysis and identify patterns, class distributions, and relationships among clause topics, lengths, and ratings.
3. To preprocess and restructure the clause data appropriately for machine learning and NLP models.
4. To engineer and evaluate features such as clause embeddings, clause length, legal keyword presence (e.g., "waive," "sole discretion," "without notice"), topic category, and readability metrics.
5. To develop and compare suitable supervised machine learning and NLP classification techniques.

6. To evaluate the models using appropriate metrics such as accuracy, precision, recall, F1-score, and confusion matrix.
7. To identify a model that provides effective classification performance while placing special emphasis on recall for the “risky” class, since missing a harmful clause defeats the purpose of the tool more than an occasional false alarm.

3 Potential Domains to Explore

The project primarily falls under the following domains:

Machine Learning — Supervised classification techniques will be used to distinguish between risky, neutral, and safe clauses.

Natural Language Processing — Clause text will be processed using NLP techniques such as embeddings, keyword extraction, and readability analysis.

Consumer Protection & Legal Technology — The project explores the application of computational methods to automated legal document review and digital literacy.

Data Science — Data preprocessing, exploratory data analysis, feature engineering, visualization, and model evaluation will form important parts of the project.

Explainable Machine Learning — Feature importance and model interpretability may be explored to understand which words, phrases, and topics contribute most to a clause being flagged as risky.

4 Dataset

- **Dataset Name:** ToS;DR Policies Dataset (Raw)
- **Source:** ToS;DR (Terms of Service; Didn’t Read), via Zenodo
- **DOI:** <https://doi.org/10.5281/zenodo.15012282>
- **Format:** Five relational CSV files (`cases.csv`, `points.csv`, `topics.csv`, `services.csv`, `documents.csv`), joined via shared ID keys
- **Total Raw Volume:** 286.5 MB across all files
- **Number of Instances (training rows):** Individual clauses in `points.csv` (≈ 17.3 MB, estimated 15,000–20,000 quoted clauses after preprocessing)
- **Target Variable:** classification (from `cases.csv`, joined onto each clause via `case_id`)
- **Classes:**
 - good $\rightarrow$ Safe
 - neutral $\rightarrow$ Neutral

– bad / blocker → Risky

- **Missing Values:** Present in some optional fields (e.g., `wikipedia`, `related`, `annotation_ref`); handled during preprocessing.

File-Level Column Breakdown

`cases.csv` — ≈ 480 rows (one row per distinct legal issue/case type). Columns:

- `id` — unique case identifier
- `classification` — good / bad / neutral / blocker (this project’s target label)
- `score` — severity score, 0–100
- `title, description` — human-readable case summary
- `topic_id` — links to `topics.csv`
- `privacy_related` — boolean flag
- `docbot_regex` — keyword/regex pattern historically used to auto-detect this case in text
- `created_at, updated_at` — timestamps

`points.csv` — estimated 15,000–20,000 rows (one row per quoted clause; this is the primary training table). Columns:

- `id` — unique point identifier
- `title` — short label, usually mirrors the case title
- `quote_text` — the actual clause text extracted from the document (**main input feature**)
- `case_id` — links to `cases.csv` for the risk label
- `service_id` — links to `services.csv`
- `document_id` — links to `documents.csv`
- `source` — URL of the source document
- `status` — approved / declined / pending / changes-requested / disputed / draft
- `quote_start, quote_end` — character offsets of the quote within the source document
- `analysis, rank, old_id, annotation_ref, service_needs_rating_update` — auxiliary/legacy fields
- `created_at, updated_at` — timestamps

`topics.csv` — ≈ 60 rows (one row per broad category, e.g., Data Collection, Content Ownership). Columns:

- `id, title, subtitle, description, created_at, updated_at`

`services.csv` — several hundred rows (one row per online platform, e.g., Facebook, Spotify). Columns:

- `id`, `name`, `url`, `wikipedia`, `keywords`, `related`, `slug`
- `is_comprehensively_reviewed` — boolean flag
- `rating` — overall service rating aggregated from its cases
- `status`, `created_at`, `updated_at`

`documents.csv` — 267.8 MB, full source ToS/Privacy Policy text per service (used for context/segmentation reference, not row-level training). Columns:

- `id`, `name` (e.g., “Privacy Policy”, “Cookie Policy”), `url`, `xpath`
- `text` — the full raw document text
- `service_id`, `reviewed`, `status`, `crawler_server`, `created_at`, `updated_at`

For this project, the primary training table is built by joining `points.csv` → `cases.csv` on `case_id` (to attach the risk label) and `cases.csv` → `topics.csv` on `topic_id` (to attach the topic category). This produces one row per clause with columns: `quote_text`, `classification` (target), `topic_title`, and `service_id`. Restructuring clause-level samples across many platforms in this way helps the model generalize rather than overfitting to a single document. Additional engineered features to explore include clause embeddings, clause length, legal keyword presence, and readability metrics.

The dataset will require cleaning before use, since raw crowd-sourced text can contain missing quotes, inconsistent topic labels, non-approved status points, or duplicate entries across services, which will be handled as part of preprocessing.

5 Potential Machine Learning Approaches

The following approaches may be explored and compared as part of the project:

- Logistic Regression
- Naive Bayes
- Support Vector Machine (SVM)
- Random Forest
- Gradient Boosted Trees (e.g., XGBoost)
- Transformer-based text embeddings (e.g., BERT/Sentence-BERT) with a downstream classifier
- Other suitable classification techniques, if required

The final selection of models will depend on the results obtained during exploratory analysis and experimentation.

6 Model Evaluation

Since the problem is a multi-class classification task (risky, neutral, safe), model performance will be evaluated using multiple metrics, including:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

7 Expected Outcome

The expected outcome of the project is a comparative analysis of different machine learning and NLP classification techniques for ToS/Privacy Policy clause risk flagging. The project aims to determine which approaches can effectively distinguish risky, neutral, and safe clauses using the available clause text and metadata.

The analysis is expected to provide insights into:

- The characteristics of the dataset and its class distribution across topics and services.
- The most informative features and keywords for identifying risky clauses.
- The effectiveness of different machine learning and NLP algorithms for legal text classification.
- The trade-offs between different evaluation metrics, particularly recall versus precision for the risky class.
- The suitability of machine learning as a computational aid for improving user awareness and digital literacy around legal agreements.

The resulting models will be considered as analytical and educational tools to help users quickly spot clauses that warrant closer reading, rather than a substitute for professional legal advice.

8 Dataset Reference

ToS;DR (Terms of Service; Didn't Read). ToS;DR Policies Dataset (Raw) — 21/07/2023 [Dataset]. Zenodo.

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